

The cached dyadic machine (v0): heads at a distance under Kraft — rev 2

Rev 2 (2026-08-09, after Mathijs’s corrections): rebuilt on the convo’s actual model ([55]–[59]), which is sharper than rev 1’s fringe sketch. Rev 1’s “moves inside the fringe are free” mechanism is dropped — the convo prices differently and better.

1. The problem

B4: bridge the multi-head machine to a machine with caches, making fast state a priced structural resource. The convo already built most of this ([56]: Mathijs’s distance-indexed cursor proposal; [57]: the Kraft analysis); this AIP freezes the v0 slice to mechanize.

2. The model (from the convo, with indices)

- **Down fills, up evicts.** A down at level ℓ morally fills the level- ℓ cache; up evicts it (Mathijs 2026-08-09; the amortization-as-normalization view of [21] — rewrite down·up $\rightarrow \varepsilon$ — is the same fact equationally). Eviction is therefore *in* the base model already; D’s walk IS the cache traffic.
- **Per-level price $c(\ell)$.** The base machine prices every level 1 ($c \equiv 1$ — the honest-information price, our current D). The physical variant scales per level: $c(\ell) = 2^{(\alpha\ell)}$ with $\alpha = 1/2$ gives random access $\Theta(\sqrt{N})$ — the “memory latency goes as the square root of capacity” lore as a stated model variant, not a patch.
- **Heads at a distance, Kraft budget** ([56]/[57]): every head has a distance d ; the constraint $\sum_i 2^{(-d_i)} \leq 1$ (Kraft) says only so much fits nearby. The heads’ fast-state shares — head i gets fraction $2^{(-d_i)}$ of the caches — form a prefix-free code, i.e. an antichain of mount slots in the fast-state block. The physical picture: a head’s share is an aligned window, cache-line-style (power-of-2 floor boundaries, not centered — Mathijs). The formal mechanics: every move of a head at distance d pays a surcharge, cost $1 + d$ — its address depth in the implicit fast-state tree ([57]; no tree is walked, the distance *is* the address). Within-stream amortization survives with the multiplicative $(1 + d)$.
- **No overlapping heads** (Mathijs 2026-08-09): shares are disjoint (the antichain condition); no dynamic cache-sharing optimization — that would make everything non-local, and [57] records why half-measures there are treacherous (adversarial sharing breaks classic bounds). Static disjoint shares keep every bound provable and sound.
- **Size-aware distance refinement** ([57] correction 1, noted for v1): a depth- d head’s state costs $\sim d$, so the honest constraint is $\sum s_i \cdot 2^{(-d_i)} \leq S_{\text{fast}}$ — the (grade, distance) antichain. v0 mechanizes pure Kraft; the size refinement falls out when heads are literally mounted in a fast-state block.

3. Theorem targets

1. *Streaming at a distance*: the scan/zip/copy bounds survive with factor $(1 + d)$ — e.g. zip at $d = 2$ per head (Kraft: $3 \cdot 2^{(-2)} = 3/4 \leq 1$, the convo’s own worked example [59]).
2. *Entropy bound* ([57]): touch frequencies p , distances $d_i = \lceil \log 1/p_i \rceil \rightarrow \text{cost} \leq n \cdot (H(p) + 1)$; matching Shannon/Kraft lower bound: no distance assignment beats the empirical entropy per

touch. (Precedent: biased search trees, Bent–Sleator–Tarjan; adaptive distance assignment via dynamic Huffman reweighting is v3.)

3. *The Kraft-vs-cliff witness* ([57]’s honest retraction, upgraded to a theorem): k -way merge under Kraft distances costs $n \log n$ for every k — the tradeoff cancels exactly, so the model provably does NOT award the external-memory multi-way win. This is the sharpest statement of the missing eviction/capacity axis, and proving it keeps us honest: Kraft pricing never under-charges ([938]: pessimism is the right failure mode for a verifier).
4. *$\sqrt{\cdot}$ -latency variant*: under $c(\ell) = 2^{\ell/2}$, random access costs $\Theta(\sqrt{N})$ and streaming stays amortized $O(1)$ -per-leaf up to the level-price sum — the physical-lore pricing as theorems.

4. Remaining taste calls

1. (was rev 1’s q1/q3/q4 — all resolved by Mathijs: aligned window picture; eviction = up, in-model; down fills, no separate refill charge.)
2. **Level-price scope**: mechanize the $c \equiv 1$ distance model first and state the $\sqrt{\cdot}$ -variant (target 4) desk-level, or mechanize the general $c(\ell)$ machinery at once? — rec: $c \equiv 1$ first; general $c(\ell)$ is a re-parameterization of one lemma layer, cheap to add after the distance layer exists. (This replaces rev 1’s “two-axis tower” question, which dissolves: the Kraft budget is normalized to 1, no separate fast-memory grade in v0.)
3. **[72] stays open, now precisely**: zip’s write head — equal *touch* frequency says $d = 2$ like the readers ([59]); double *data* rate says a bigger share. The tension is exactly touch-frequency distances vs size-aware distances (§2 last bullet), so the answer belongs to the size-aware refinement, not to v0 guesswork. — rec: record, defer to the size-aware distance refinement in v1.

4b. The dirty-up refinement (Mathijs musing, 2026-08-10 — open)

With free up (AIP-2 amendment), down = fill and up = commit: a *clean* up is a free discard, a *dirty* up (writes below since the descent) would pay a write-back and propagate dirtiness to the parent. Machinery: a dirty marking on nodes; total write-back cost of a written region $\approx$ its size — sound. Payoff worth weighing: under dirty-up pricing, zip’s write head pays its ups while the read heads don’t — the read/write asymmetry of real hierarchies falls out of the base machine, and **[72] may resolve here rather than in the distance layer** (“writes twice the data” becomes “pays write-backs”), no Kraft choice needed. Pricing principle behind the whole family: free is admissible exactly for potential-consuming operations (up consumes depth); read/write/ down can repeat in place and must stay paid. Status (updated 2026-08-10): *direction endorsed* by Mathijs — the cocycle reading (Expo 3) stays categorically clean and yields cohomology links; open design questions collected in `aips/accepted/0005-dirty-cocycle.md`; pricing details remain his call at call 2. The unconditional-free-up retrofit (read-only fiber) is in flight on `track/free-up-refresh`.

5. Sequencing

Mechanization track (after Mathijs reads this rev): the $1 + d$ cost on action words + Kraft feasibility + distance-aware streaming (target 1), then entropy bound (target 2), then the cliff witness (target 3). Independent of the expositions/receipts line. Targets 1–2 are DONE (2026-08-10; Expo 2).

Decided 2026-08-10 (Mathijs): after the $C = 1$ identity paragraph, do the Euclidean variants *immediately* — $c(\ell) = 2^{(\ell/2)}$ and $2^{(\ell/3)}$ ($k = 2, 3$) as first-class variants next to $c \equiv 1$, not a distant v1 nicety. Their entropy story is Campbell/Rényi (shannon-programme §2); folklore anchors ibid.